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Write a pseudocode for a greedy approximation algorithm for Set Cover. Clearly define inputs (universe, subsets) and output (cover set). Design algorithm selecting subsets with maximum coverage. Use loops and selection logic. Ensure clarity in pseudocode. Handle all cases. Justify completeness.

Code:

```
def set_cover(U, S):
    cover = []

    while U:
        best = max(S, key=lambda x: len(x & U))

        if not (best & U):
            return "No Complete Cover Possible"

        cover.append(best)
        U -= best

    return cover

U = {1, 2, 3, 4, 5}
S = [
    {1, 2, 3},
    {2, 4},
    {3, 4, 5},
    {5}
]

print(set_cover(U, S))
```

Output:

```
[{1, 2, 3}, {3, 4, 5}]
```

Design a pseudocode for approximating Maximum Cut in a graph. Clearly identify inputs (graph) and output (partition). Develop algorithm assigning vertices to partitions. Use loops and condition checks. Ensure correctness of approximation. Present structured pseudocode. Handle all cases.

Code:

```
def max_cut(g):
    A, B = set(), set()

    for v in g:
        a = sum(1 for u in g[v] if u in A)
        b = sum(1 for u in g[v] if u in B)

        if a > b:
            B.add(v)
        else:
            A.add(v)

    return A, B

graph = {
    0: [1, 2],
    1: [0, 2, 3],
    2: [0, 1, 3],
    3: [1, 2]
}

A, B = max_cut(graph)

print("Partition A =", A)
print("Partition B =", B)
```

Output:

Partition A = {0, 3}

Partition B = {1, 2}